Android Demo Script

26 October 2026 | 15-minute live sequence and 5-minute video plan.

Use the real working search as the main demonstration. Lab 4 permits prioritizing main functionality when implementation time is limited. Do not present planned features, prototype operator data or widget fixtures as live backend behavior.

1. Preflight

Check	Action
Device	Use the actual Android demo phone. Unlock it, enable USB debugging and confirm adb lists it as device, not unauthorized.
Services	Start the model on laptop port 8001 and API on 8000. Inspect model artifact health and run a genuine eligible search; health alone is insufficient.
USB	Run adb -s DEVICE_ID reverse tcp:8000 tcp:8000. The APK targets phone loopback port 8000; reconnecting USB may require reverse again.
Network	Laptop needs database/OneMap/HDB access. Phone needs internet for OneMap tiles and external navigation. Keep the USB cable connected.
Freshness	Rehearse and check metadata, tariff review dates and current source readings. Refresh/review through the normal owner workflow if necessary; do not disable guards for the demo.
Scenario	Choose a real current 15-90-minute arrival example and a short-trip example. Live availability and winners may change. Keep a second rehearsed destination.
Privacy	Use public sample places in recordings. Keep .env files, passwords and keys off screen; entering operator credentials should be obscured.
Backup	Record and review the five-minute screen video before the presentation. The script is prepared; the actual video has not been recorded by this task.

2. Live demo - 15 minutes

Time	Segment	Actions / explanation	Role
00:00-01:00	Problem and users	Explain that the driver arrives later than the latest reading. Introduce anonymous driver search and the separate operator area.	Driver presenter
01:00-02:00	Android input	Show the actual device and one foreground GPS fix if available. For a repeatable forecast scenario, manually select NTU as the origin and Jurong East MRT as destination.	Driver presenter
02:00-05:00	Real recommendation	Select a stay duration and Best overall. Explain driving ETA to each carpark, forecast probability of at least one lot, current source timestamp, supported total-stay cost and walking route. Read actual values; do not quote fixed expected counts.	Driver presenter
05:00-06:00	Preference trade-off	Re-search using Prioritise cheaper and Prioritise closer. Explain that weights change while eligibility rules do not. A different winner is not guaranteed with every live shortlist.	Driver presenter
06:00-07:00	Map and navigation	Show OneMap attribution and start/destination/carpark markers. Open external navigation for one carpark, then return. No physical driving is needed.	Driver presenter
07:00-08:30	Short-trip fallback	Use a rehearsed short route such as Novena to City Hall. Explain the disabled sub-15-minute forecasting cohort and the separate current-reading list if fresh usable evidence exists.	Driver presenter
08:30-09:30	Recovery and scope	Show a rehearsed recoverable input/provider error. If time permits, show theme persistence or tracker UI, clearly labelling non-durable/prototype behavior. Do not present mock operator data as live monitoring.	Driver presenter
09:30-11:30	Traceability	Trace UC-01 through RecommendationController, pricing helpers, model adapter and the CTL/REQ tests. Show one overnight tariff flow graph and its actual expected/observed table.	Technical presenter
11:30-13:00	Engineering evidence	Summarize automated tests, independent basis paths, unsupported-price handling and prediction/current separation. Distinguish interface/regression checks from model accuracy and on-phone evidence.	Technical presenter
13:00-14:30	Architecture and extension	Explain Flutter -> API -> DB/OneMap/model; collection is separate. Mention canonical IDs, observation/knowledge time, prospective forecast auditing and private server credentials.	Technical presenter
14:30-15:00	Close	State implemented search benefits and remaining scope. Leave the app on a useful result screen. Keep the pre-recorded video available for network failure.	Driver presenter

3. Phone commands and recovery

From the repository root, use the setup in backend/README.md. The debug build permits local HTTP; it is not the production deployment configuration.

adb devices -l
adb -s DEVICE_ID reverse tcp:8000 tcp:8000
adb -s DEVICE_ID install -r lab4/submission/ParkChoiceSG-debug-arm64.apk
adb -s DEVICE_ID shell am start -n com.example.parkchoicesg_2/.MainActivity

If this happens	Recovery
No device / unauthorized	Unlock the phone, accept debugging, verify the USB data cable, then list devices again.
Phone cannot reach API	Confirm backend is running; rerun reverse for the selected device. Do not substitute the laptop LAN address without rebuilding the matching API URL.
Location denied or slow	Use manual place selection. Refresh a fix older than two minutes; do not invent a coordinate.
No eligible forecast	Explain the actual exclusion. Try the rehearsed alternative; current-only fallback is valid only when the backend permits it.
Price unavailable	Explain missing/stale/ambiguous tariff evidence. Do not quote a made-up hourly price or turn missing cost into zero.
Provider/network outage	Show recoverable state, then use the reviewed recording clearly identified as pre-recorded evidence.

4. Five-minute recording script

Time	What to capture
00:00-00:30	Problem, actual phone and anonymous search.
00:30-01:15	Select real origin/destination and duration.
01:15-02:30	Show a real forecast recommendation; explain ETA, probability, supported price and walking distance.
02:30-03:15	Re-search with a different ranking mode; show actual trade-offs.
03:15-04:00	Open map and external navigation; return safely.
04:00-04:40	Show short-trip current fallback or an honest unavailable state.
04:40-05:00	State the tested scope and close on the result screen.

Lab 4 asks for a five-minute video; Lab 5 permits five to seven minutes. A five-minute recording fits both. This plan does not assert that a video file exists. Capture the actual Android app and check picture, audio and readable labels before adding the real video/link to the submission index.